

Linguists are Exceptional Inductive Reasoners – A Case Study on Skill Relevance in Enterprise Software

This 20-minute talk will follow a case study of my work as a linguist on an engineering team developing a generative AI product for business workflows. Linguists have a uniquely strong background in analysis on imperfect and incomplete data. Linguistic analysis necessitates considering small samples of data to make and test hypotheses about a language as a whole. In the tech world, and particularly enterprise software, there is a heavy emphasis on large amounts of data and scaling. The prevailing attitude is that human annotation, analysis, and evaluation is too costly, time-consuming, and small scale to be worth the investment. As part of an engineering team working to ship a generative AI product, I was able to leverage linguistic expertise and demonstrate value throughout the data pipeline from training data to evaluation.

- Training and Annotation IS LIKE Developing Elicitation Plans

Linguists at ServiceNow often work with third-party contractors to manage data annotation projects. Crucial skills include (1) understanding technical goals, (2) communicating these goals to non-expert labelers, and (3) designing projects to efficiently collect the most data. The first two skills are often understood as strengths of linguists, but I found the third most impactful and directly analogous to linguistic research. Planning data collection for any linguistic research, especially fieldwork, necessitates a detailed strategy to get the right data, get enough data, and make necessary trade-offs. My content expertise is not in enterprise workflows, but breaking down an annotation project by feature, volume, variation and time apply regardless of topic.

- Evaluation IS LIKE Writing Problem Sets

My team relies on a single numerical metric calculated on a golden dataset to evaluate the product and quality of various models. This is extremely common in the world of machine learning and AI. The product takes in natural language input and has a structured workflow output (Béchar and Marquez Ayala 2024). Due to privacy policies, the dataset needs to be created by hand. My coworker and I took on a months-long overhaul of the golden dataset. This type of work had not been done by the engineering team, partially for time constraint reasons, and partially because of concerns about overfitting to a test set. We discovered and described 3 major issues, all of which were supported by our analytical linguistic background. First, there were actual errors, where the natural language input and structured output did not match. Developing the guidelines for labelling the training data made us adept at finding these cases, which are not always obvious. Second, there were many cases of ambiguity, where the natural language prompt did not have a 1:1 mapping to the output and had multiple realistic outputs. Ambiguous inputs also need to be evaluated from a product perspective, but in a small dataset with an exact match metric, these can have a sizeable impact on the evaluation metric. Third, we found the distribution of features in the evaluation dataset did not align with the priorities for the product. Some high priority features were rare or absent, and some unsupported features were disproportionately present. We took all these insights and proposed changes, deletions, and new data for the golden dataset which we were able to implement. This undertaking involved writing and rewriting prompts/outputs to capture a representative distribution of features and ensuring the dataset and metric were calibrated to be maximally informative of performance. We needed to be experts in both the natural language prompting side as well as the structured technical output – an overlap we uniquely filled amongst teammates and stakeholders. We also had to strategically develop a small amount of data that was representative of the landscape of capabilities for a versatile project. The process was remarkably similar to writing problem sets as a linguistics instructor.

- Error Analysis IS LIKE Doing Morphophonological Analysis

Recently, I have taken on the role of error analysis on the team. My approach is bottom up, looking into individual samples to identify model behavior, then assessing particular problems across the dataset. In addition to adding interpretability to the evaluation metric, my understanding of the whole pipeline allows me to identify likely sources and possible solutions to model behaviors. Certain issues can be more easily resolved with training data, others with prompt engineering, and still others with model training or changes to the retriever. Going between detailed looks of small amounts of data and a bigger-picture system understanding is a skill essential to linguistic research that is invaluable in a business context.

Direct contributions to the product informed by linguistic expertise include: (1) planning, writing guidelines, training, and validating annotation projects; (2) creating ground truth data that accurately represents product goals and aligns with predetermined metrics; and (3) interpreting model performance and proposing changes to training data, prompting, and model architecture to improve the product. Although I tied each of these to a specific skill I gained from a graduate education in linguistics, the point is much more general. Linguists understand the value of inductive reasoning, of spending time on the details, of small datasets, and of outliers. In a business world driven by big data, projects at scale, and perceived efficiency of LLMs for all data work, the distinctive approach and skills of linguists can make a tremendous impact. Linguistic skills are directly applicable to a variety of business and software projects, even when the data is not in the same representation (phones, morphemes, sentences, etc.) that the researcher is trained in. In this talk, I aim to demonstrate concrete impacts my linguistic expertise had in a highly specific enterprise software product context that I had no previous content background in to show how a ‘linguistic approach’ can be leveraged in unexpected ways.